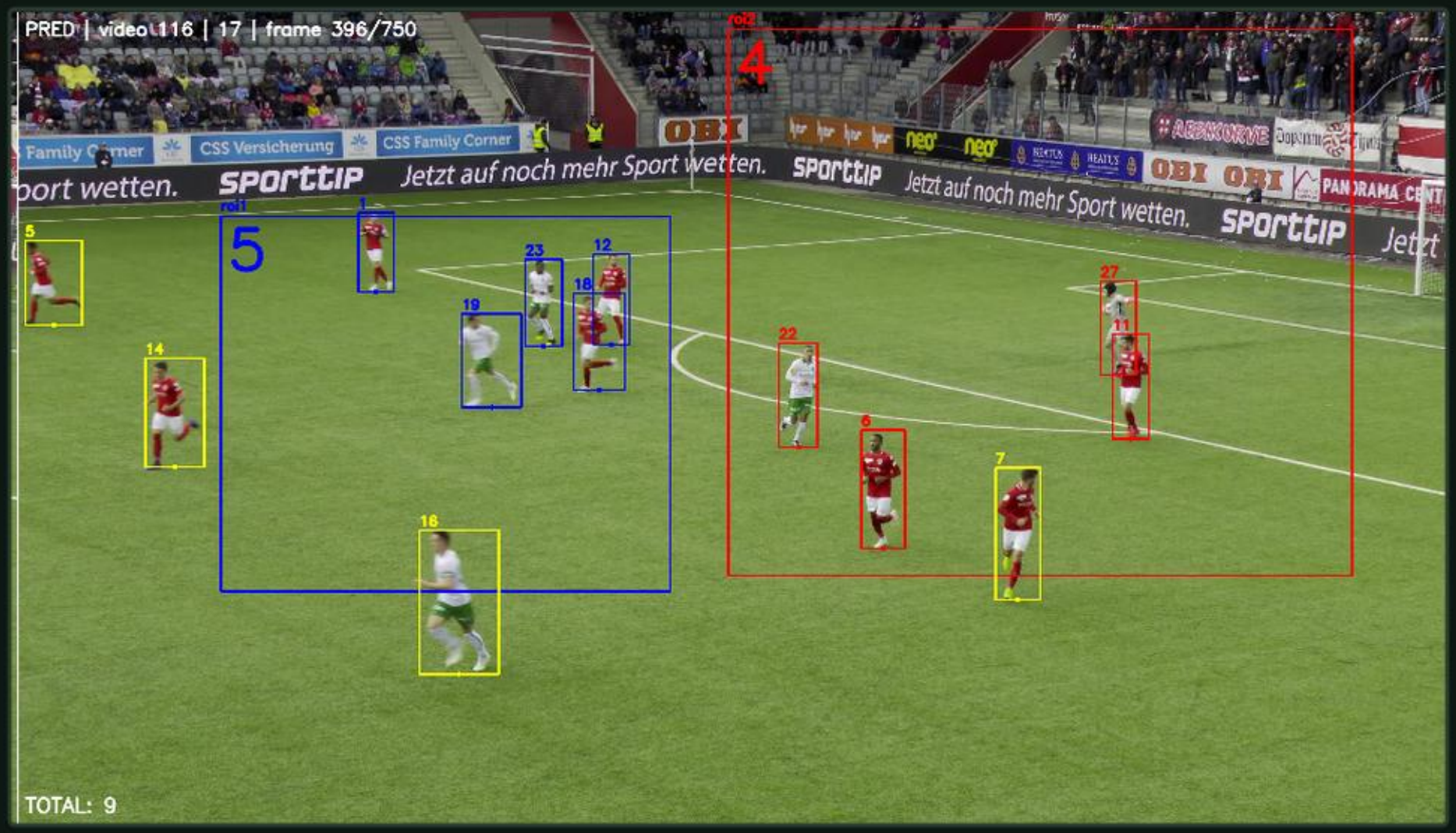
COMPUTER VISION PROJECT

Who's who on the pitch?

Tracking every player in broadcast football video, with a camera that never stops moving.



82.4%
HOTA@0.5

+20 pts
association acc.
vs. baseline

1.807
final score
(PTBS, max 2)

THE PIPELINE

Four stages, one job: never lose track of a player.

1 Detect

YOLOv8m fine-tuned on SoccerNet

Backbone frozen, fitness on mAP@50, 1280 px input

2 Track

BoT-SORT + camera motion compensation

Short-term continuity frame to frame

3 Describe

OSNet Re-ID embeddings

Each player crop becomes a 512-d appearance vector

4 Remember

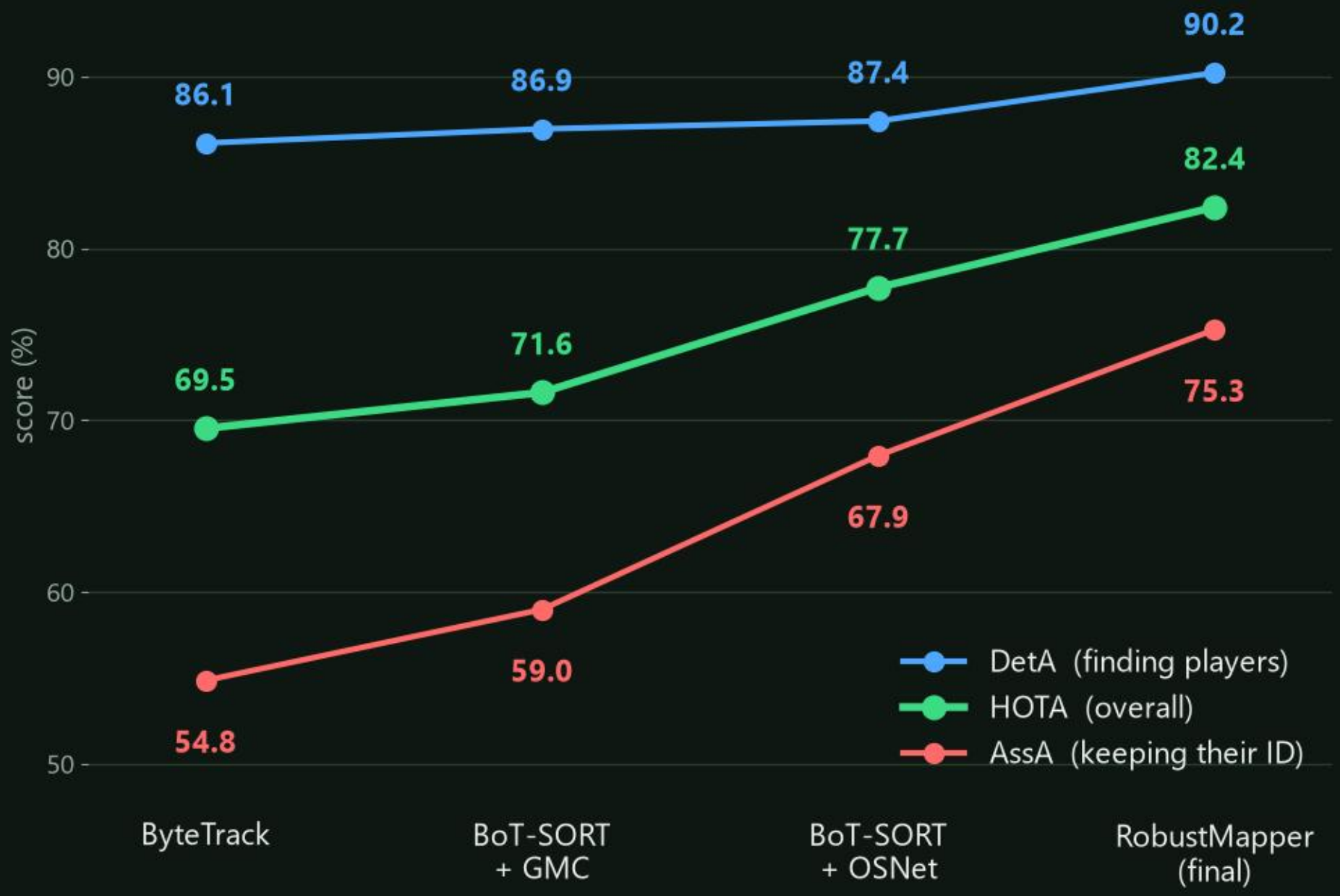
RobustMapper (ours)

Gallery memory, probation, merge & recovery of lost IDs

Output per frame: player boxes with persistent IDs
+ how many players stand inside each region of interest.

WHAT ACTUALLY MOVED THE NEEDLE

Detection was already good. Identity was the hard part.

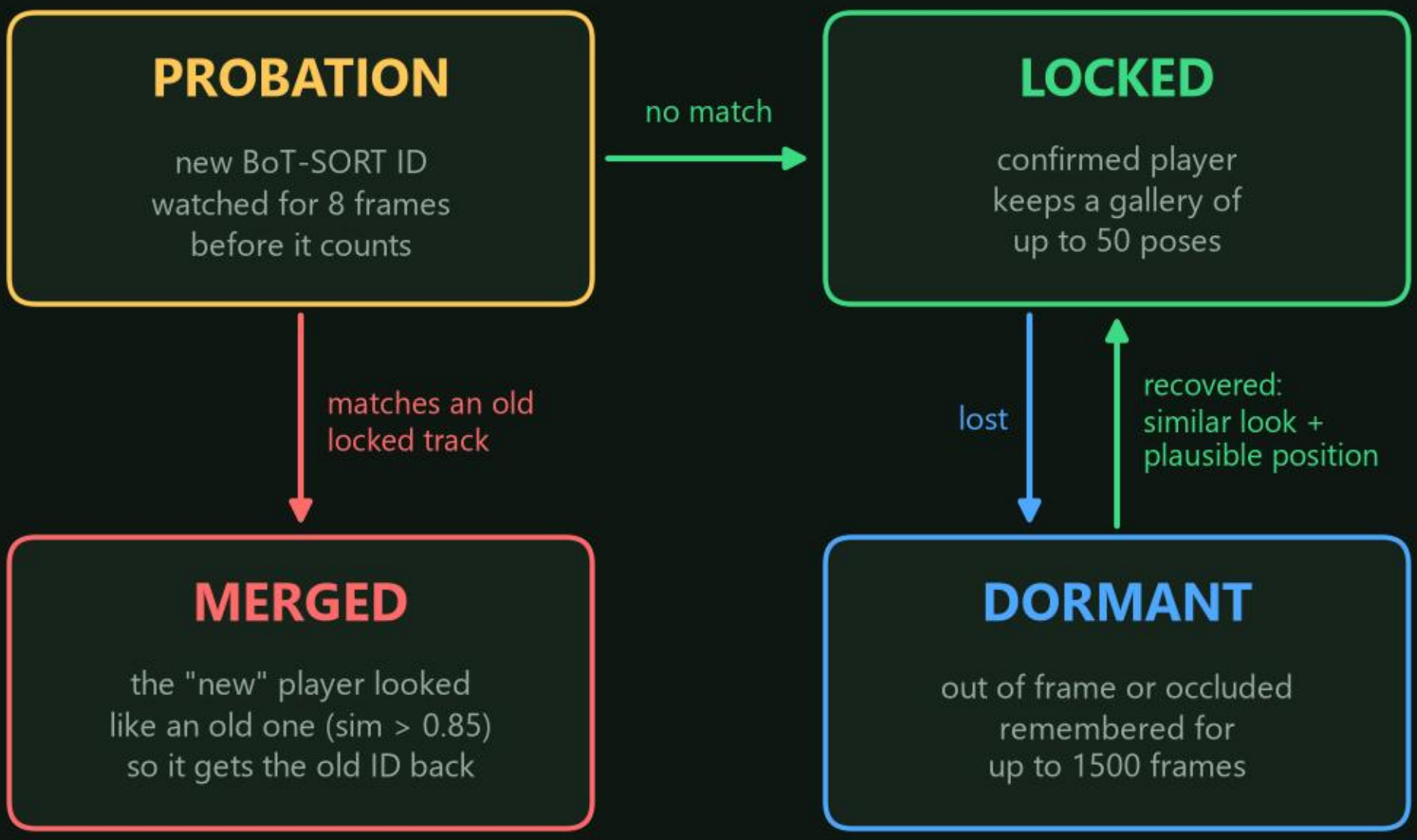


AssA: 54.8 → 75.3

Most of the gain came from remembering players, not from a better detector.

INSIDE ROBUSTMAPPER

Don't trust a new ID until it earns it.



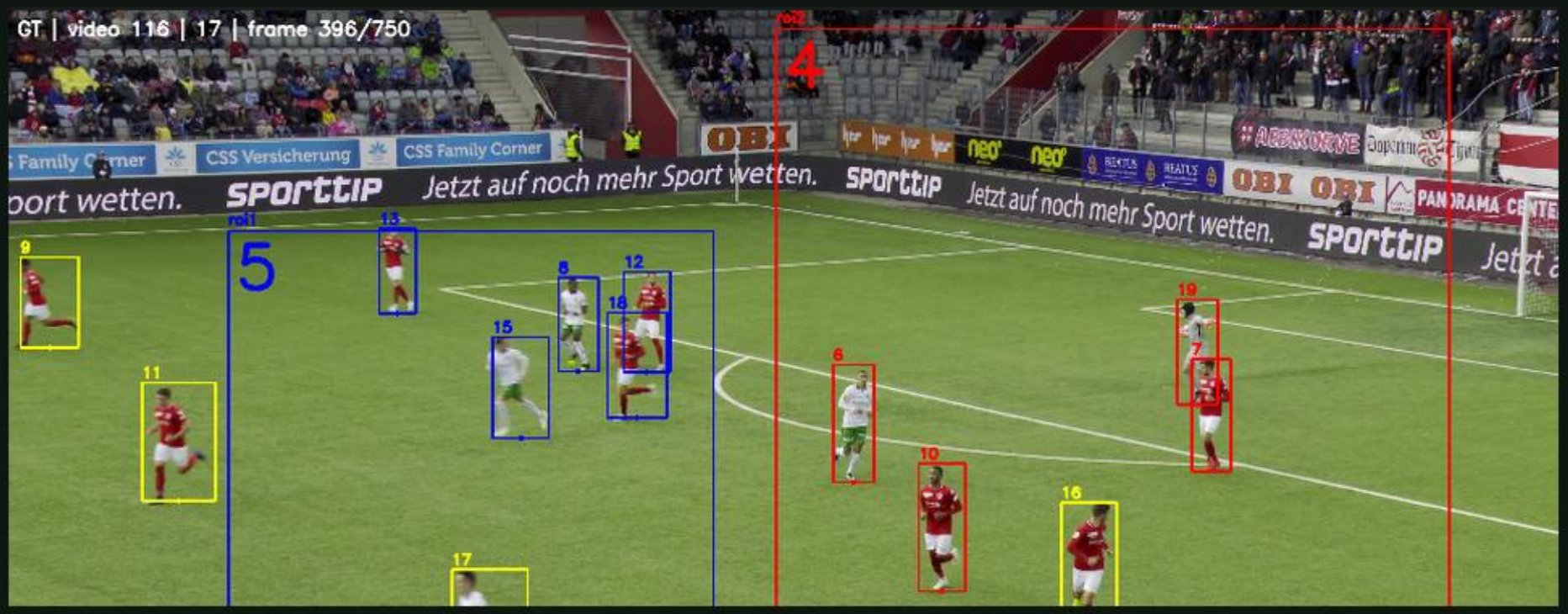
Why a gallery and not a running average?

A player seen from the back, from the side, and half-hidden behind a teammate looks like three different people. Keeping several poses instead of one averaged vector made re-identification after occlusions much more reliable.

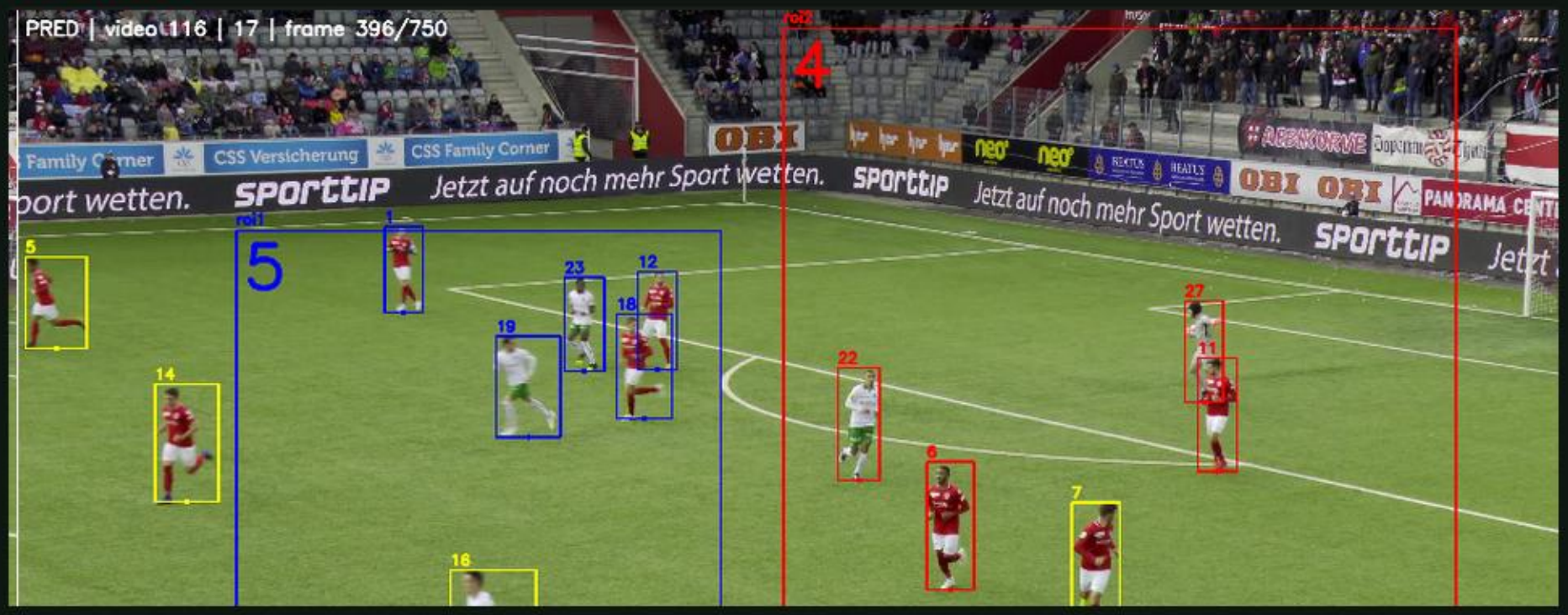
RESULT

Ground truth vs. our output

Ground truth



Prediction



Players counted in the blue and red regions: 5 and 4, same as the ground truth.

82.4	90.2	75.3	0.983
HOTA	DetA	AssA	nMAE